

Maths for Computer Science

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1 Introduction

This worksheet is intended to remind you of some basic principles of mathematics that you are likely to have come across before you started your university course. Many of the ideas you come across here will be useful when we start to look at programming in Python.

We will start with some simple mathematical notation, before looking at some algebra. We will then look briefly at Vectors before moving onto graphs and some of the intuition behind calculus.

This notebook is intended to accompany the taught lecture and provides a number of questions with spaces to write in your answers to check your understanding.

2 Mathematical notation

Let's start with the most basic of mathematical notation, addition, subtraction, multiplication and division. I know this is very basic, but we're going to build up from here.

Question 1.

$$\begin{array}{rcl} 12 + 20 & = & (1) \\ 1.2 + 2.05 & = & (2) \\ 23.5 - (-2.7) & = & (3) \\ 2 \times (1 + 5) & = & (4) \\ 7 \div 21 & = & (5) \end{array}$$

Next let's have a look at the power notation.

Question 2.

$$\begin{array}{rcl} 2^2 & = & (6) \\ 2^3 & = & (7) \\ 2^0 & = & (8) \\ 2^{-1} & = & (9) \\ 2^{-2} & = & (10) \\ 2^{0.5} & = & (11) \end{array}$$

One of the things that is useful to know when dealing with number like this is that we can apply little tricks.

Question 3.

$$2^3 = 2 \times 2 \times 2$$

$$2^2 = 2 \times 2$$

so

what is $2^3 \times 2^2$?

what is $2^3 \div 2^2$?

Can you write a description of how multiplication and division of powers can be done quickly?

There are lots of little tricks like this that you can use in manipulating mathematics. There is no hidden secret here, but like any language you get more comfortable with it the more you use it.

3 Using letters instead of numbers

One thing that confuses people when they first start doing maths is the use of letters instead of numbers. But using letters allows us to generalise problems and link the numbers we use to real world problems. This is also going to be key when we start looking at writing computer programs to tackle problems.

The circumference of a circle is equal to twice the radius times **pi**. **pi** (written π) is an example of using a symbol in mathematics to represent a number. In this case π the ratio between the radius of a circle (the distance from the middle to the edge in a straight line) and the circumference (the distance all the way around the circle). It is approximately 3.1416 but when we write it as a symbol everyone knows what it represents and we don't need to be approximate.

We will also use a letter to represent the circumference, c seems sensible and one for the r for the radius. To write the **equation** for the circumference of a circle we write

$$c = 2\pi r \tag{12}$$

A few things to note here ...

- I haven't said what value c or r are yet just that this relationship between the values exists.¹
- I haven't included a $\times$ symbol
- I haven't put spaces between the symbols

Question 4.

Using the equation given above calculate the circumference of a circle with diameter 3 cm?
How should you round your answer to 2 decimal places?

A final question before we move onto some more algebra.

¹Note: when I use letters in place of numbers in an equation (as I did with c and r I will refer to them as variables.

Question 5.

What is the volume of a sphere? Write your answer in mathematical notation and then calculate the volume of a sphere with a radius of 1.7, giving your answer to 3 decimal places. You can find the equation from the internet.

4 Algebraic expression

An *algebraic expression* can be a single number, single variable or a collection of numbers and variables which are connected by the basic arithmetic operations (+, −, ×, ÷) and powers. For example all of the following are arithmetic equations:

$$2ax - 3by + z \quad (13)$$

$$5x^2 + 3x - 2 \quad (14)$$

$$\frac{4}{3x+2} + \frac{2x-3}{2x+2} \quad (15)$$

$$\sqrt{x^2 + y^2} \quad (16)$$

Often the equation we are given is not in the simplest form possible and we will want to simplify it.

Question 6.

Think about each of these expressions, is there a simpler way to write them?

$$2x^2 + 3x + 4 + x + 3x^2 + 2 \quad (17)$$

$$\frac{60a^2b}{35a^5c} \quad (18)$$

5 Equations

An equation is any statement in which two expressions are placed on either side of an = sign to indicate that they are equal.

In maths we might use an equation to say that both sides of the expression are equal like this:

$$x^2 - y^2 = (x - y)(x + y) \quad (19)$$

Or we might use it to show how one thing is dependent on another, like this:

$$A = \pi r^2 \quad (20)$$

where A is the area of a circle and r is the radius of a circle.

Finally we might use an equation to define a condition on a problem, like this:

$$2x + 1 = 21 \quad (21)$$

Question 7.

Often we want to be able to "solve equations" to find the value(s) of a variable which satisfy the equation. Find the value(s) of x which satisfies the following equation:

$$x^2 - 2 = 0 \quad (22)$$

List 5 values of a and b which satisfy the following equation:

$$2a + b = 100 \quad (23)$$

When we have an equation which features two variables one way of understanding what the equation is telling us is to visualise. Let's consider the following equation:

$$y - 6 = x^2 - 4x - 2 \quad (24)$$

The easiest way to start this is to rearrange the equation so that one of the variables is on its own. In this case let's make y be the left hand side of the equation by adding 6 to each side.

$$y = x^2 - 4x + 4 \quad (25)$$

We could simplify this if we want to be:

$$y = (x - 2)(x - 2) \text{ or } y = (x - 2)^2 \quad (26)$$

Next we need to define a range of values of x for which we want to know the value of y . Let's assume we want to see how y changes for $0 \leq x \leq 4$ in steps of 0.5:

x	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	4	2.25	1.0	0.25	0.0	0.25	1.0	2.25	4.0

Having worked out the values of y for each value of x we can now plot the values on a graph which we show in Figure 1.

Sometimes we can say something about a function before we even plot it out. For example I could tell you that the value of the function is 0 at $x = 2$.

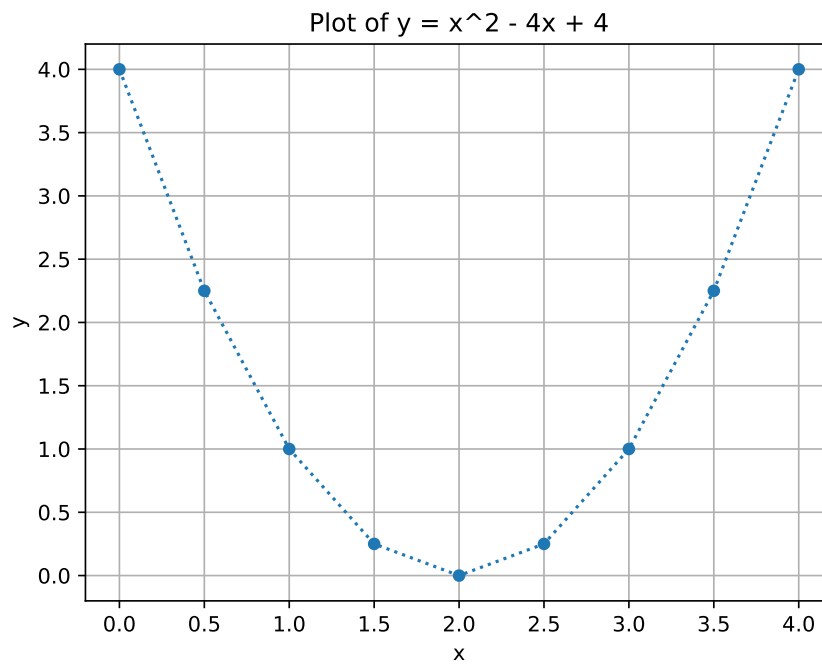


Figure 1: Graph of the function $y = (x - 2)^2$

Question 8.

At what value of x will this function have a value of zero? **Note:** we call these values the roots of the equation.

$$y = (x + 3)(x - 2) \quad (27)$$

Of course people have spent a lot of time writing tools which make working with mathematics and algebra easier, so lets have a look at a tool which helps us to visualise functions.

Visit <https://www.geogebra.org/calculator>.

6 Vectors

In this section we are going to start looking at a method of representing data with multiple values. For example we might want to think about the points we calculated in the previous exercise, and which we represented on our graph. Here each data point has two values, it's position with respect to the x-axis and its position with respect to the y axis. So we might have a point we could represent each point as:

$$\begin{bmatrix} x \\ y \end{bmatrix}$$

Specifically, for Figure 1, we might say:

$$d_1 = \begin{bmatrix} 0.0 \\ 4.0 \end{bmatrix}, d_2 = \begin{bmatrix} 0.5 \\ 2.25 \end{bmatrix} \dots$$

This representation of numbers in a column format is what we call a vector. Storing information in this way allows us to do some nice mathematical tricks. For example we can add to vectors together like this²:

$$\begin{bmatrix} 1.0 \\ 4.0 \\ 2.5 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 2.0 \\ 3.0 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 6.0 \\ 5.5 \end{bmatrix}$$

This way of looking at sets of numbers is really useful for computer science as computers are often optimised to execute functions on vectors in a more efficient way than doing each function one at a time for every value.

Question 9.

What is the answer to wech of the following

$$\begin{bmatrix} 1.0 \\ 4.0 \\ 2.5 \end{bmatrix} + \begin{bmatrix} 1.0 \\ 4.0 \\ 2.5 \end{bmatrix} = \quad (28)$$

$$\begin{bmatrix} 1.0 \\ 4.0 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 2.0 \\ 1.0 \\ 1.5 \end{bmatrix} = \quad (29)$$

$$2 * \begin{bmatrix} 1.0 \\ 4.0 \\ 2.5 \end{bmatrix} = \quad (30)$$

6.1 Vectors forms

One of the things we are often interested in computer science is how two things are related and vectors are good way of doing that.

Let's consider the distance between York and Leeds we might think of this on a map as shown in Figure 2. Here then the distance east is approximately 22.4 km and the distance north is 17.8 km so we could write this as a vector as follows:

$$\begin{bmatrix} 24.4 \\ 17.8 \end{bmatrix}$$

²Notice that these vectors have 3 numbers, what might they represent?

Notice that one thing we have done here is associated meaning with each element of the vector. We know that the first element is the distance travelling east and the second element is the distance travelling north.

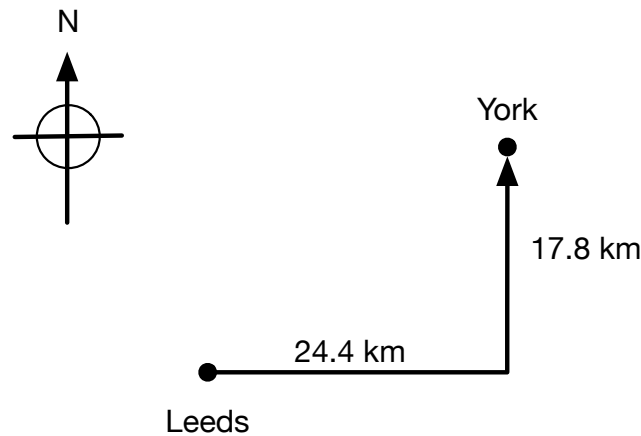


Figure 2: The distance from Leeds to York

Question 10.

Looking at Figure 2 above what does the following vector represent?

$$\begin{bmatrix} -24.4 \\ -17.8 \end{bmatrix}$$

The representation we have used so far is called the component form. Another way of representing this data is in geometric form. This form states the values as a **magnitude** (length of a straight line connecting the two points) and the **direction** (angle at which it is drawn).

Consider then Figure 3 where the vector has magnitude 30.2 km and the direction N36°E so that $v = (30.2, 36^\circ)$.

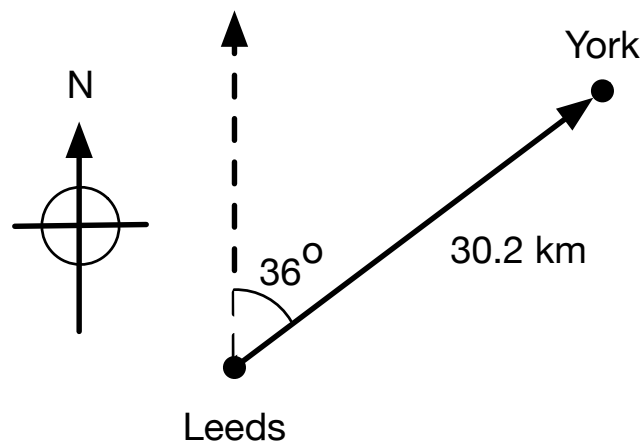


Figure 3: The distance from Leeds to York in geometric form

Note : There are well defined rules for moving between vector forms and I leave that as an exercise for the reader. Note that the correct form to use depends on what we are trying to do with the vectors.

Why does this matter?

Vectors are used to represent all kinds of problems in Computer Science and sometimes we want to think about how different (or similar) things are so given two vectors we might want to get a **measure** of closeness.

Alternatively we might want to know what happens when one process, represented by a vector, is followed, or acted on by another.

So how might we work with vectors? We don't have time to go through everything we use vectors for, but let's consider a simple example. Think of a routing problem for a vehicle, each route is made up of multiple segments and each segment can be represented by a vector. If we know the distance from point A to point B and we know the distance from B to C then we can combine the vectors to tell us the distance from A to C.

Consider then the following points in a 2 dimensional space.

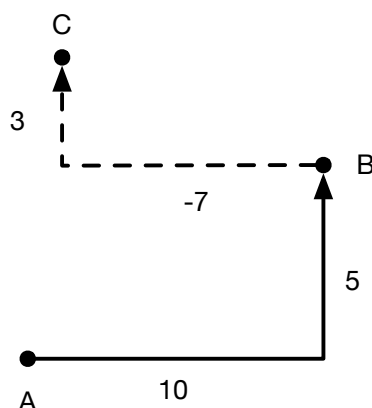


Figure 4: Vector Addition

We can represent this with 2 vectors.

$$AB = \begin{bmatrix} 10 \\ 5 \end{bmatrix} \text{ and } BC = \begin{bmatrix} -7 \\ 3 \end{bmatrix}$$

Question 11.

So what is the value of the vector AC if $AC = AB + BC$?

Another common example of how vectors are composed is when trying to calculate the cumulative force acting on a body this is a common problem in physical robotics.

Note: Addition is only one form of composition we can apply other functions to vectors but we leave that to the reader to explore in their own time.

Whilst this all look pretty simple when we have only two route segments, notice that exact the same procedure could be used if we were adding up 100,000 different vectors (or a thousand vectors each with 100 elements). Indeed this is where computers become very useful because it's a pretty simple bit of maths done very many times.

Let us consider a journey from York to london where every individual segment of road is recorded as a vector. We could write the distance between the start and end point of the journey by adding together all the vectors on the route:

$$v_1 + v_2 + \cdots + v_n$$

where v_i is a vector containing two numbers and n is the number of journey segments undertaken. Another way we might write this mathematically is:

$$\sum_{i=1}^n v_i$$

This is an example of how maths can start to look very complicated but for those who speak maths it makes the writing of complex operations much simpler. For example, don't worry about the greek letter here $\sum$ called sigma just means sum of and the values below and above it mean for all values of i in the range 1 to n . This is like lots of jargon it's intended to make it easier for the expert to write (and read) but can be daunting for the non-expert. Usually there is a simple English language explanation for the intention of a mathematical term.

Indeed we could rewrite this in other forms for example:

$$\sum_{i=1}^n v_i = v_1 + v_2 + v_3 + \dots + v_n$$

Even this form uses a mathematical notation $\dots$ but I really don't want to write out a thousand vectors, it feels like a waste of time!

So far we have thought about vectors with only two numbers. This has allowed us to print the visualisation of them on the page which is useful, but we aren't limited to two numbers.

6.2 Data encoding and Bigger Vectors

So the vectors we've considered so far are pretty simple but we make extensive use of vectors in computer science to encode data from the real-world into a form that the computer can deal with. For example, computers don't understand the concept of chair or cold they are simply mathematical computation devices so we need to convert concepts we want to work with into a mathematical form and, in vectors, we have ways of doing that. For example we could all agree that

$$1 = \text{cold}, 2 = \text{warm}, 3 = \text{hot}$$

Question 12.

How might you encode the following pieces of data for use in calculations within a computer?

1. A health record data for a patient
2. An ECG (heart beat recording)
3. A colour
4. A photo of a cat

In each case we will have a process which maps each piece of information in the input record to an element in the vector. We can then undertake calculation on the vector to process or analyse data within a record or to compare two records.

Note: Although we have thought about vectors of numbers (This is a Maths recap after all) when we start looking at computers you will notice that we can have collections of more complex types, imagine a vector for which each element is a vector!

7 Matrices

A Matrix can be thought of as a set of vectors which are collected together into one data structure. Let's think about a problem of solving simultaneous equations. Consider the following pair of equations:

$$x + 2y = 4 \quad (31a)$$

$$3x - 5y = 1 \quad (31b)$$

Now we could use equation manipulation to solve this but instead let's convert the set of x and y coefficients into vector for that is:

$$(1, 2), (3, -5)$$

such that the first element is the coefficient of x and the second is the coefficient of y .

Now if we stack the two vectors on top of each other we get a matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}$$

So how is this helpful? Well by combining matrices and vectors we can write the two equations we had earlier as a single equation like this³:

$$\begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

or

$$A.X = C$$

where A is a matrix containing the coefficients we have, X is a vector containing the variables we want to solve for and C is a vector of the right hand values from the equations⁴.

You might be able to work out from the way I've written this how matrix multiplication works. We work along each row in the matrix and down the vector. Let me demonstrate in the general case.

$$\begin{array}{c} \longrightarrow \\ \left[\begin{array}{cc} v_{11} & v_{12} \\ v_{21} & v_{22} \end{array} \right] \left[\begin{array}{c} x \\ y \end{array} \right] \downarrow \end{array} = \begin{bmatrix} v_{11} \times x + v_{12} \times y \\ v_{21} \times x + v_{22} \times y \end{bmatrix}$$

Question 13.

Multiply the matrix by the vector to get the values of a and b .

$$\begin{bmatrix} 2 & 4 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

³Note that the shape of the brackets is not important, sometimes we use round brackets sometimes they are curved, it varies by text book.

⁴Note that I have used the $\cdot$ (dot) operator here to indicate matrix multiplication rather than a traditional scale multiplication we will see how this is different shortly.

7.1 Vector Transformations

One interesting use of Matrices is it “transforming” vectors that is leaving them the same shape but moving the point that they represent to a different position in the vector field⁵.

Let’s take a look at Figure 5. Here we see four vectors (A, B, C, D) we can think of the relationship between them as a rotation, that is that if we rotate A counter clockwise around the origin we get vector B, rotating twice gives us C etc. If you’ve ever used Photoshop (other graphics packages are available) then this is a common transformation.

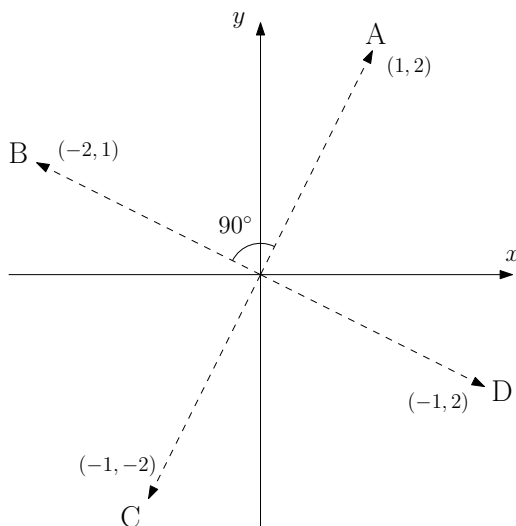


Figure 5: Rotating a vector using a Matrix

So how might we do this with a Matrix? All we need to do is construct the correct matrix and multiply the matrix by our vector.

Question 14.

Multiply the following Matrix by the vector A to confirm it is a rotation resulting in the vector B.

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} =$$

Repeat the exercise for vector D to show the matrix returns us to the vector A.

More generally we can define our rotation matrix as:

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

where θ is the angle (in radians) we want to rotate the vector.

This is only one type of transformation and as anyone who has used a graphics package will know there are many others (shift, scale, skew) but this idea of applying a matrix transform is commonly used.

⁵a fancy way of saying to another vector which shares the fundamental features of the original vector.

It's also worth noting that this doesn't have to be in 2 dimensions. It might be harder to visualise an n dimensional transformation but the idea that we are mapping it to a different position in the feature space still holds.

7.2 Element wise operations

Just a quick note before we move on that it is possible to do element wise operations on matrices.

If we want to multiply or divide a matrix by a scalar we simply apply the multiplication (or division) to each element in turn so we can take the following matrix and divide by 11 to get:

$$\frac{1}{11} \times \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} \frac{5}{11} & \frac{2}{11} \\ \frac{3}{11} & \frac{-1}{11} \end{bmatrix}$$

Indeed can also multiply matrices in an element wise manner, the only restriction is that the matrices are of the same shape. So

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \odot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 9 & 16 \end{bmatrix}$$

Notice here that we use a strange symbol to indicate this is an element wise multiplication $\odot$. It's important to use the correct symbol to avoid confusion.

7.3 Matrix multiplication

More usually however when we are using matrices in mathematics we use the rows times columns method and we can extend this to multiply two matrices. Consider the following:

$$\begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix} \cdot \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} \times \frac{1}{11}$$

If we consider the second matrix to be simply two column vectors side by side then we can calculate the product of the two matrices as

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix} \cdot \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} &= \begin{bmatrix} 1 \times 5 + 2 \times 3 & 1 \times 2 + 2 \times -1 \\ 3 \times 5 + -5 \times 3 & 3 \times 2 + -5 \times -1 \end{bmatrix} \times \frac{1}{11} \\ &= \begin{bmatrix} 11 & 0 \\ 0 & 11 \end{bmatrix} \times \frac{1}{11} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

This matrix with 1's on the diagonal and zeroes everywhere else is a special matrix. It's called the identify (I) and it works a like the number 1 in normal maths. So if we multiply any matrix by I we get the same matrix out.

Question 15.

Multiply the following two matrices together to confirm you get the same matrix as you started with:

$$\begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

So how does this apply to our first example in this section of simultaneous equations?

Well remember how we wrote the matrix equation our as:

$$A.X = C$$

Well we can rearrange this using the same “language” we would use for any other mathematical equation, i.e. we divide each side by A to get X on its own.

$$X = A^{-1}.C$$

This new matrix A^{-1} is the inverse of A and is the matrix such that $A.A^{-1} = I$. Just like the inverse of 4 is 4^{-1} such that $4 \times 4^{-1} = 1$.

Lucky we have already seen what the inverse of A is and we have shown that $A.A^{-1} = I$:

$$A^{-1} = \frac{1}{11} \times \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix}$$

So let's complete our solution of the simultaneous equations

$$X = \frac{1}{11} \times \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \times 4 + 2 \times 1 \\ 3 \times 4 + -1 \times 1 \end{bmatrix} / 11 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

So let's just double check our answer in equation 1 from our set of simultaneous equations $x + 2y = 4$.

if $x = 2$ and $y = 1$ the $2 + 2 \times 1 = 4$ which is correct.

7.4 Getting the computer to help

So how do you calculate the inverse of a Matrix? Well I leave this to the interested reader as a research exercise, but in the meantime let me show you how to get the computer to do it for you.

1. First visit Wolfram Alpha <https://www.wolframalpha.com>
2. Click on the math input button below the text input
3. enter the text “Inverse()”
4. move inside the brackets then click the **small** button with looks like two brackets with 4 dots inside it.
5. Select the large 2 by 2 matrix button
6. Enter the matrix values into the boxes that appear
7. Press the equals button to evaluate your query

Although we have used the graphical buttons to enter matrices this time we can also enter them as curly brackets. First make sure we are in Natural Language mode then enter the following text into the wolfram alpha text prompt:

```
{{2, -1, 0}, {-1, 2, -1}, {0, -3, 4}} {{x}, {y}, {z}} = {{0}, {-1}, {4}}
```

The Curly brackets are around not only the matrix, but each row. Notice that the line we have entered isn't just a matrix but a set of simultaneous equations in matrix forma and wolfram will give us the solution.

Question 16.

Where possible calculate the following if the calculation is not possible say why.

1. $[1, 2] \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} =$

2. $\begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot [1, 2] =$

3. $[1, 2, 3] \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} =$

A note in closing: Vectors and Matrices are at the heart of efficient computation especially in AI. Whilst it is possible to doing one calculation after another this gets very time consuming on large problems. Solving problems with Matrices gives us the opportunity to undertake many calculations in a more efficient manner even undertaking calculations in parallel.

8 Understanding Graphs

One of the first steps in understanding data is often to visualise it (as we did earlier). In Figure 6 I have plotted some data which we recorded from a car. What can you tell me about the car from this graph?

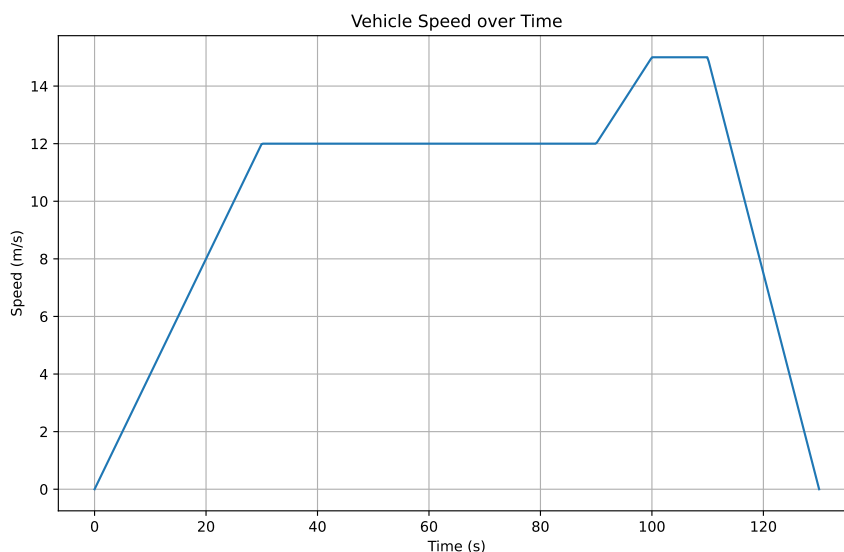


Figure 6: Graph of vehicle speed over time

Question 17.

Looking at the vehicle speed graph in Figure 6 how would you calculate the :

1. maximum speed experienced by the passenger
2. maximum acceleration experienced by the passenger
3. distance travelled by the vehicle

These might look like contrived questions (and they are) but they introduce a couple of really important mathematical concepts.

8.1 Rate of change

One of the things we are often interested in data is how quickly things are changing. Let's take a look at another graph, a straight line, Figure 7.

Here we have plotted the straight line given by $y = \frac{1}{2}x - 1$. To calculate the rate of change (or gradient) for this line we need calculate the amount of change in y for the amount of change in x . Or to write this mathematically:

$$\frac{dy}{dx} \quad (32)$$

This is due short hand for “difference” in y divided by difference in x . Another way we might see this written is :

$$\frac{d}{dx} 0.5x - 1 \quad (33)$$

since $y = 0.5x - 1$.

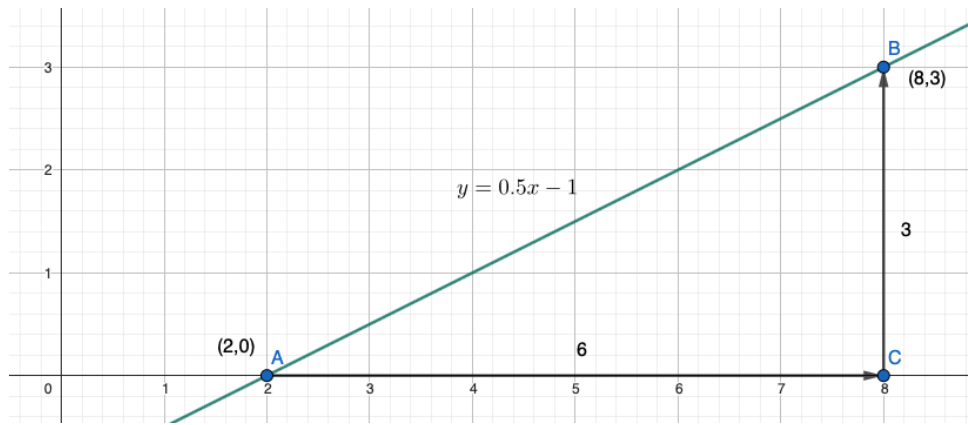


Figure 7: Gradient of a straight line

So looking at our graph we can see that if we travel from point A to point C we move 6 in the x axis and 3 in the y axis so

$$\frac{dy}{dx} = \frac{3}{6} = 0.5 \quad (34)$$

Question 18.

Without plotting the graph or calculating anything what do you think the gradient of the following straight lines are :

1. $y = 3x + 2$
2. $y = 0.6x - 0.3$

So how does this link back to question 13?

Well acceleration is a way of saying "what is the rate of change of speed". Similarly speed is the rate of change of distance and indeed if you think of how we write speeds you can see this i.e. m/s. This means the number of meters we change (y axis) for every second (x axis).

Note: If we look at the units on the vehicle speed graph we are writing $\frac{m/s}{s} = m/s^2$ which is acceleration (the rate of change of speed).

Well that's all fine for straight lines but what about when our graphs aren't straight? Consider Figure 8, what is the gradient of this function ($y = x^3 - 6x^2 + 9x + 2$)?

Well we can see that the curve changes it's gradient as we progress along the x axis, so it might be better to ask what is the gradient of the curve when $x = 0$.

What we would need to do to calculate this, from the graph, is draw a straight line that at that point which just touches the graph without crossing it (a tangent), we can then simply calculate the gradient as we did before. In fact this is what we've done in Figure 9

Let's have a look at this more closely with GeoGebra:

<https://www.geogebra.org/graphing/jpzhfe8v>

We can see that the sign of the gradient tells us if the graph shows an increasing (or decreasing) trend at the point we are looking at, or indeed if the gradient is zero.

This last case is particularly important as it means we are at a local maximum or minimum for the graph.

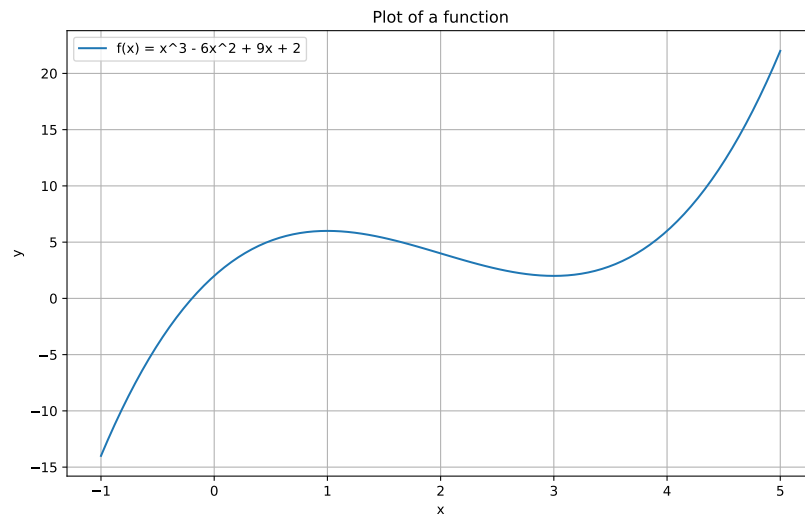


Figure 8: Polynomial Function

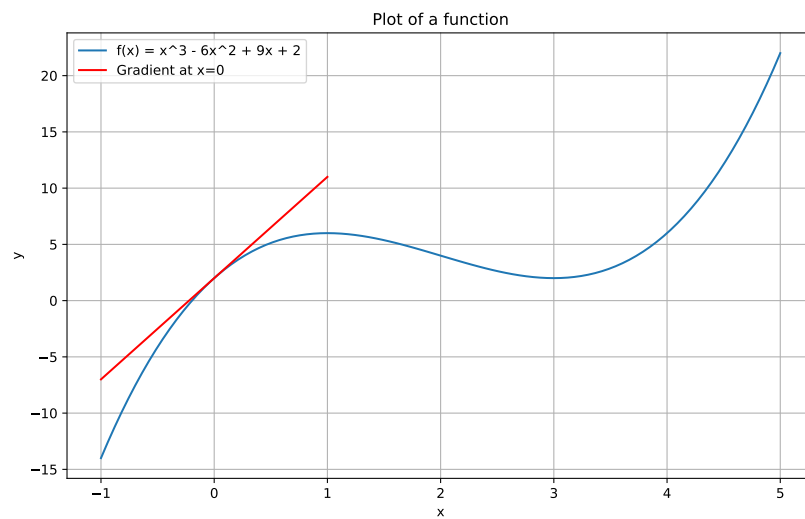


Figure 9: Polynomial function with gradient shown at $x = 0$.

Maximization (or Minimization) of a function is something that lies at the heart of AI (something you will talk about in other courses) so having the mathematical tools to allow us to do this is important even if we don't know the equation in advance.

Question 19.

Consider the following: you have a piece of code that, when you enter a value for x , returns the value of y and the gradient at that point. You might find it helps to consider the graph in Figure 9 but remember this procedure should work for any function.

Using plain English write a set of steps to find the maximum value returned by the function when $0 < x < 5$ that the function returns.

Step 1 : Choose x at random and get the gradient and output value y

Step 2 :

Well done you've just written your first algorithm!

8.1.1 Differentiation

If you want to get further into this subject you need to read about a technique called differentiation. The previous section was intended to give you the intuition of what going on, but when I say find the straight line that just touches the curve at the point you want, well that's not as simple as it seems for complex functions.

Instead there are mathematical techniques which allow us to take a class of functions and find their derivatives, i.e. a function that tells us the gradient at ALL points across the x axis.

Let us consider the function we had above.

$$f(x) = x^3 - 6x^2 + 9x + 2 \quad (35)$$

The derivative is written as⁶:

$$\frac{d}{dx}f(x) = 3x^2 - 12x + 9 \quad (36)$$

So using this new function we can calculate the gradient as

x	gradient ($\frac{d}{dx}f(x)$)
0	9
1	0
2	-3
...	...

Note: You can check these figures using the Geogebra app if you want.

Question 20.

Can you see any pattern in the functions before and after the differentiation.

⁶You'll just need to trust me for the moment that this is correct

A final note on differentiation. Whilst calculating derivatives can be complicated we live in a world where people have built tools to do much of the hard work for us. Hence we give you the intuition here not the mathematical chops for calculus (the branch of mathematics this belongs to).

Wolfram Alpha is a website which provides a great deal of mathematical tools all through a simple natural language interface.

Visit <https://www.wolframalpha.com> and enter

`differentiate  $x^3 - 6x^2 + 9x + 2$`

into the text prompt.

8.2 Area under a graph

The final thing we will cover in this brief (and pacey) recap of pure mathematics is how to find the area under a graph. In practical engineering this is often an important tool for example if we have a function of force over time then the area under the graph is the amount of energy expended.

Let's consider once more our vehicle speed graph, which we have repeated being in Figure 10.

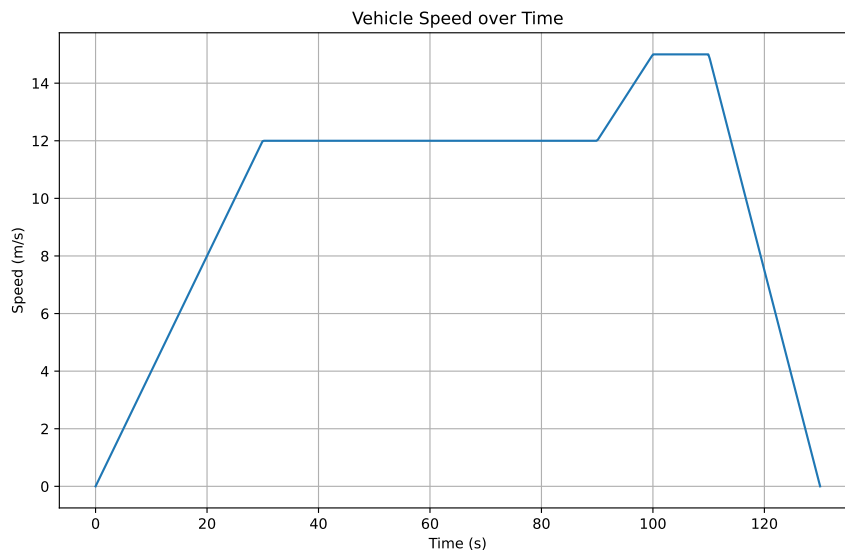


Figure 10: Graph of vehicle speed over time

We can calculate the area under the graph by splitting the graph into nice geometric shapes we understand. A triangle from time 0 to time 30, a rectangle from time 30 to 90 etc. We then add the area of each region together to get the total distance travelled.

Note: Looking again at the axis we are multiplying units so $(m/s) \times x = m$.

Not this is all fine when the function is simple shapes but what if I ask you for the area under the graph:

$$y = x^3 - 6x^2 + 9x + 2 \quad (37)$$

The first thing to note is that there isn't a limit to range of x so what do I mean by "under the graph". So let's give the are some limits let's say the area under the graph between $x = -1$ and $x = 4$.

Well if we visualise this we get the graph in Figure 11.

Our next decision is how do we deal with areas below the axis? The answer is problem specific but the default is to treat this as a "negative" area i.e. we take this away from the "positive" area above the axis line.

Without using complex maths we might try and approximate this area. We can do this by drawing a set of boxes and summing the area of all the boxes generated. Look at Figure 12. What we have done here is choose a step size (0.5 and then taken the value of $f(x)$ at each step and calculated the area of the boxes.

So here is the table of values for each point on the graph.

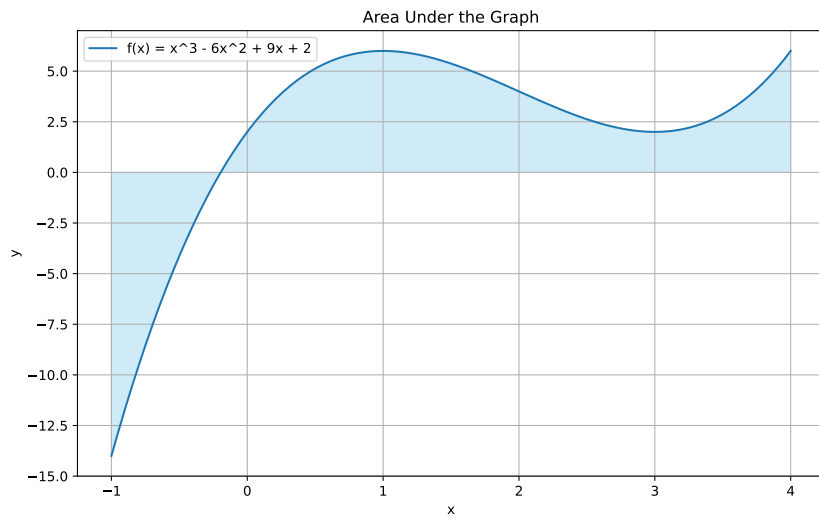


Figure 11: True Area under the graph

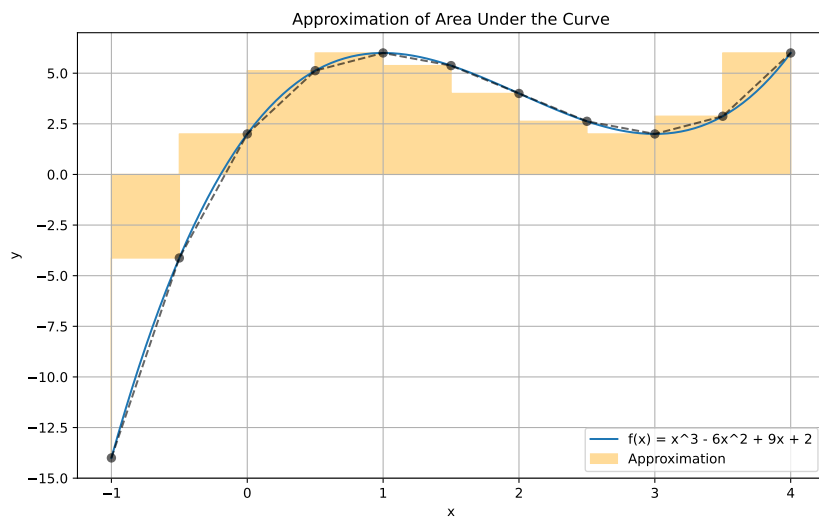


Figure 12: Approximating area using rectangles

x	-0.5	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
f(x)	-4.125	2.0	5.125	6.0	5.375	4.0	2.625	2.0	2.875	6.0

Question 21.

Calculate the area under the curve by calculating the area of the 10 boxes. e.g. the second box is 0.5 wide and 2 high.

How good is this approximation? Well we can see from our simple plot that there are boxes which are too big and some that are too small. The hope is that over these errors cancel out.

There are two ways that we can make this approximation better. Firstly we can make the step size smaller. Maybe we can try 0.1 rather than 0.5. This is shown in Figure 13.

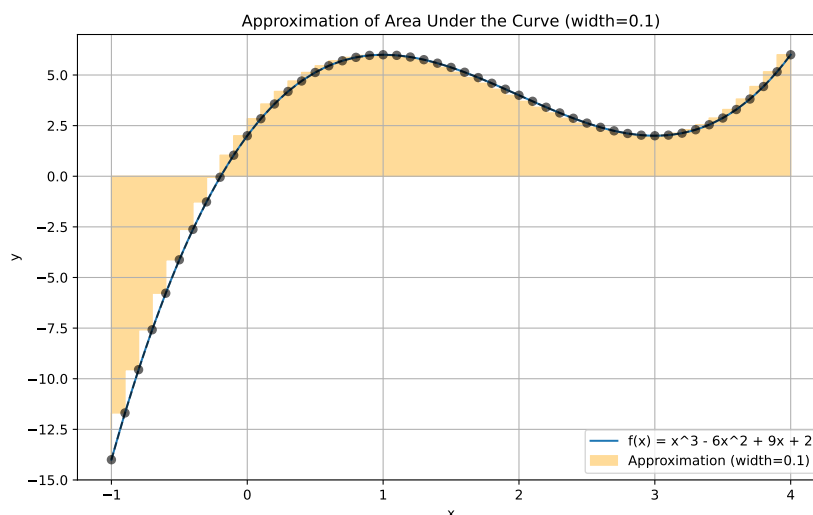


Figure 13: Approximation with a smaller step size

We can see this is much better but for the negative part of the graph we can clearly see that boxes just are good at approximating these curves.

The second method is to use a trapezoid rather than a rectangle (i.e. we put a triangle on top). This is illustrated in Figure 14.

Combining the two techniques will give us a closer approximation but it's never going to be completely accurate. To get the "correct" answer we need to turn again to mathematics and another facet of calculus, **integration**.

8.2.1 Integration

What we saw in the previous section is called numerical integration and can be applied to any function even if we can't write it Mathematically (i.e. it might just be a set of data).

When we can state the equation in a mathematical form however we can often use techniques to derive the Integral. In mathematical form we write it as follows:

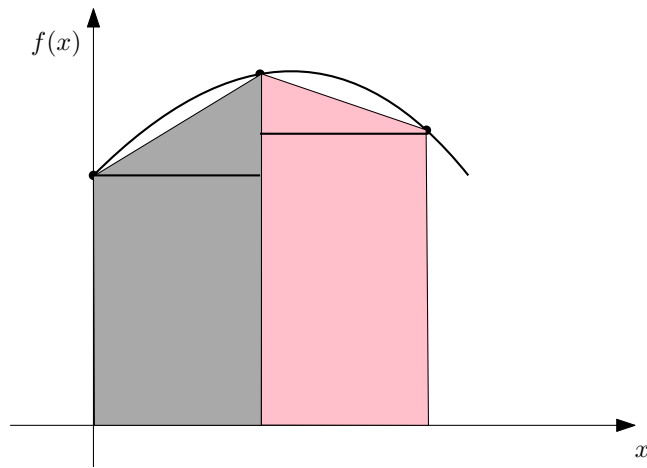


Figure 14: Approximation with trapezoids

$$\int_a^b f(x) dx \quad (38)$$

We read this as the integral of $f(x)$ between a and b with respect to x . Notice that the symbol looks like an elongated S. I always think of this as meaning sum as in "sum up all the area under the function".

For our function then we are asking for:

$$\int_{-1}^4 x^3 - 6x^2 + 9x + 2 dx \quad (39)$$

Again we don't have time to teach you how to calculate this but we do have Wolfram Alpha.

Visit <https://www.wolframalpha.com> and enter the following into the text prompt

`integrate x^3 - 6x^2 + 9 x + 2 between x=-1 and x=4`